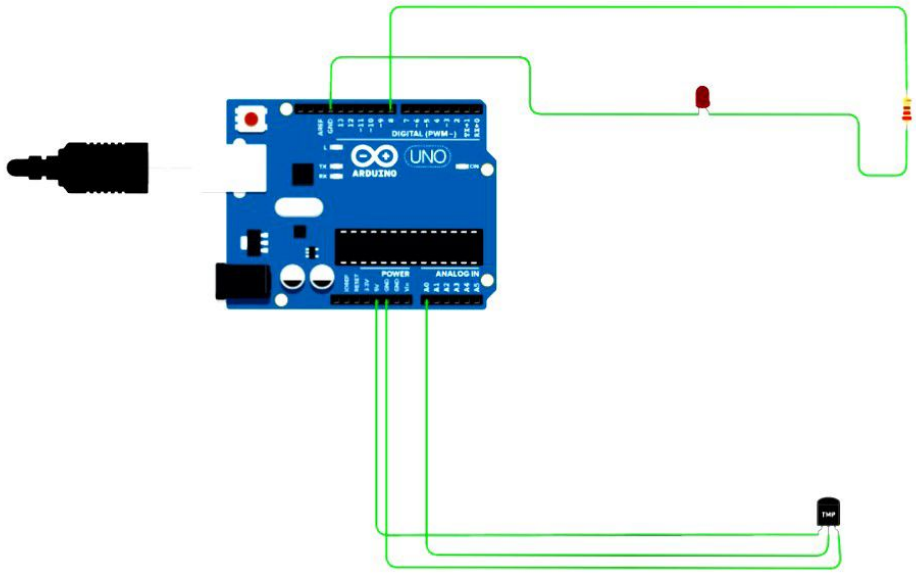


Components
Basic
Search
Resistor
LED
Pushbutton
Potentiomet...
Capacitor
Slideswitch
9V Battery
Coin Cell 3V Battery
1.5V Battery
Breadboard Small
micro:bit
Arduino Uno R3

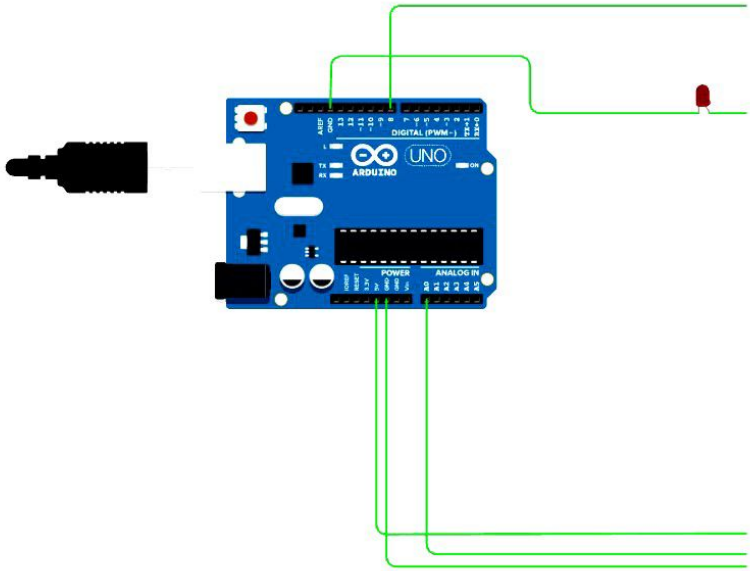
Circuit design Smooth Snicket

tinkercad.com/things/fhrmpgIVLXC-smooth-snicket/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard

Smooth Snicket

Circuits | Schematic | Components

Code | Start Simulation | Send To



Text | 1 (Arduino Uno R3)

```
const int sensorPin = A0;
const int ledPin = 8;
const float threshold = 30.0;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int sensorValue = analogRead(sensorPin);

  float voltage = sensorValue * (5.0 / 1023.0);
  float temperature = (voltage - 0.5) * 100.0;

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");

  if (temperature >= threshold) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
  }

  delay(1000);
}
```

Serial Monitor

Circuit design Smooth Snicket

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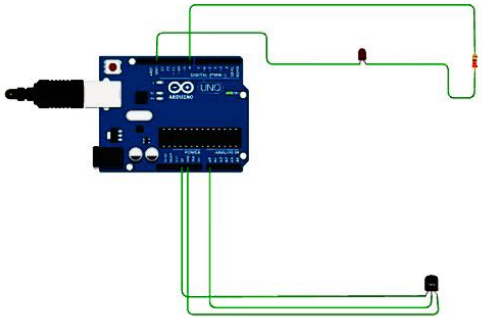
Smooth Snicket

Circuits | Schematic | Components

Simulator time: 00:00:08

Code | Stop Simulation | Send To

1 (Arduino Uno R3)



```
const int sensorPin = A0;
const int ledPin = 8;
const float threshold = 30.0;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int sensorValue = analogRead(sensorPin);

  float voltage = sensorValue * (5.0 / 1023.0);
  float temperature = (voltage - 0.5) * 100.0;

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");

  if (temperature >= threshold) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
```

Serial Monitor

```
Temperature: 24.78 C
Temperature: 24.78 C
Temperature: 24.78 C
Temperature: 24.78 C
Temperature: 24.78 C
Temperature: 24.78 C
Temperature: 24.78 C
```

Send | Clear

Circuit design Smooth Snicket

tinkercad.com/things/fhrmpgIVLXC-smooth-snicket/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard

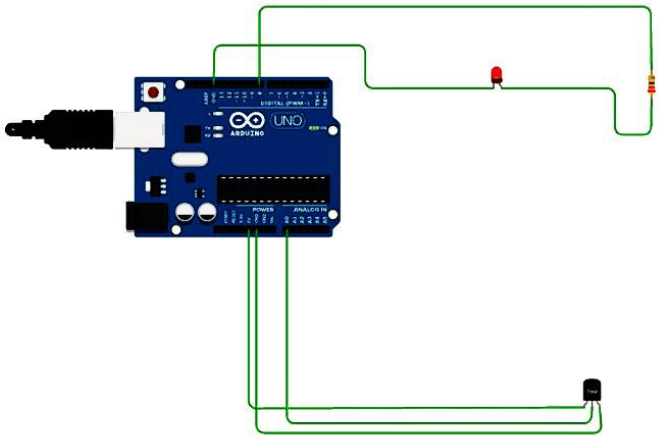
Smooth Snicket

Circuits | Schematic | Components

Simulator time: 00:01:02

Code | Stop Simulation | Send To

1 (Arduino Uno R3)



```
const int sensorPin = A0;
const int ledPin = 8;
const float threshold = 30.0;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int sensorValue = analogRead(sensorPin);

  float voltage = sensorValue * (5.0 / 1023.0);
  float temperature = (voltage - 0.5) * 100.0;

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");

  if (temperature >= threshold) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
```

How the debugger works

Add breakpoints by clicking on the line numbers.Hover over the variables while paused to see their value.Use the buttons above to resume simulation or step one line at a time.

Serial Monitor

```
Temperature: 36.02 C
Temperature: 36.02 C
Temperature: 36.02 C
Temperature: 36.02 C
Temperature: 36.02 C
Temperature: 36.02 C
Temperature: 36.02 C
```

Send | Clear